Understanding data pipelines

In this train, we'll introduce the concept of data pipelines, explaining what they are and why they are essential in data processing and workflow management.

Learning objectives

> - Understand what data pipelines are.

Introduction to data pipelines

A high-level overview of data pipelines

The term data pipeline is used to describe a set of sequential processing actions applied to data to get it from an initial state to the desired end-state. In certain applications these processing steps are independent of one another, allowing them to be executed in parallel. A data pipeline can be divided into three core concepts: the data source(s), one or more data processing tasks, and the data destination(s).

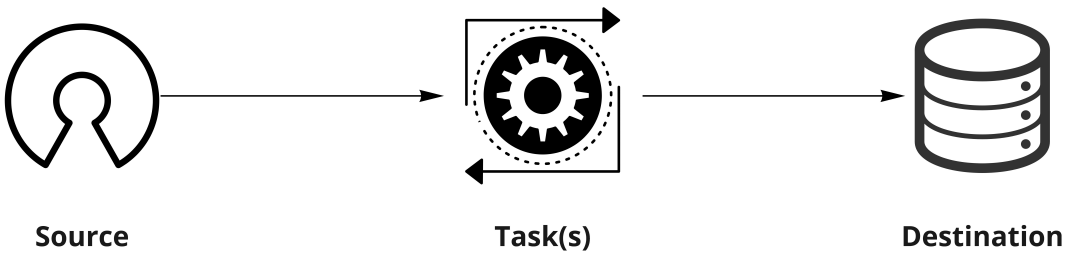


Figure 1: A high-level depiction of a data pipeline.

Data processing tasks can include: - data cleaning; - data augmentation; and - data enrichment. Let's consider a simple example of a data pipeline used to process student marks. In this example, the data source is the set of student marks for a particular exam. The end-state, or target destination, is a business intelligence (BI) dashboard used to visualise various aggregate-level and individual-level metrics. These metrics include class average, best performing student, lowest and highest scoring questions, etc. For the data to get from the source to the target in the desired format, various data transformations need to take place. For instance, we need to clean the data by checking for absent students, unanswered questions,

possible erroneous data entries, and so on. The pipeline for this specific use case might look like as shown in Figure 2.

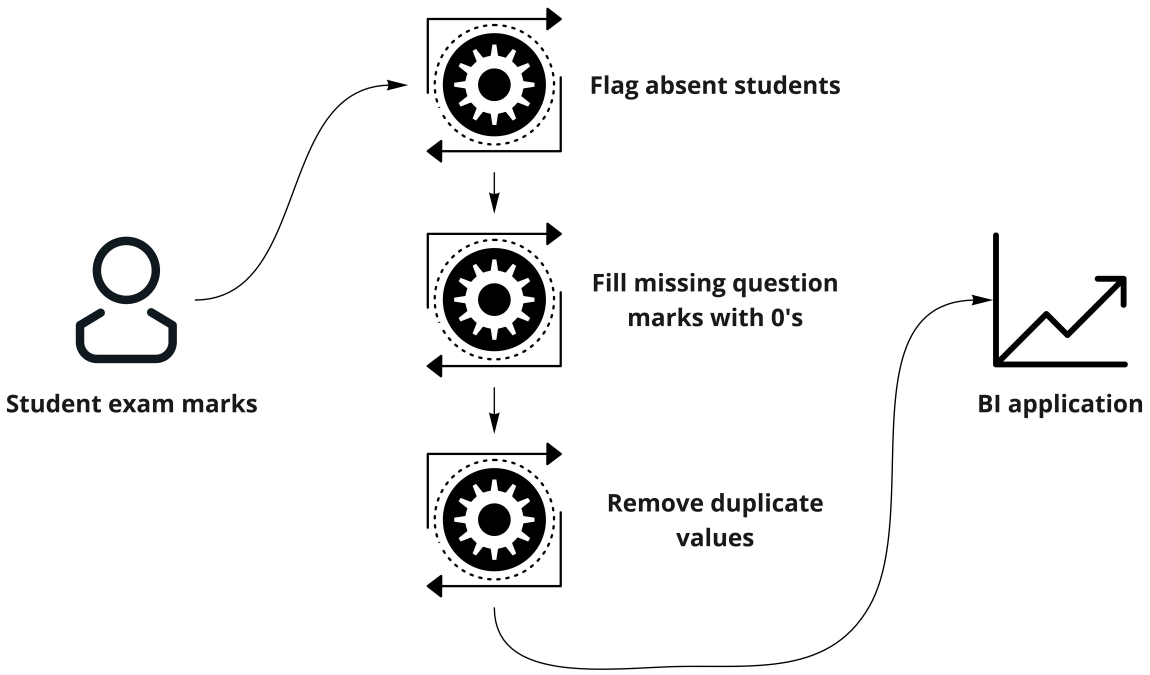


Figure 2: A depiction of our basic student exam marks data pipeline.

Conclusion

Data pipelines play a crucial role in ensuring seamless data movement, transformation, and accessibility across various systems. By structuring data processing workflows into well-defined steps, organizations can achieve consistency, automation, and efficiency in handling large datasets. Whether dealing with student exam results, business intelligence, or real-time analytics, understanding how to design and manage data pipelines is essential for any data-driven professional.

> **Refelction:** Think about how streaming services like Netflix or music platforms like Spotify provide personalized recommendations. Behind the scenes, data pipelines collect user interactions, clean the data, and feed it into machine learning models that predict user preferences. Similarly, in financial institutions, data pipelines help process transactions in real time to detect fraud. These real-world applications highlight how data pipelines are not just theoretical concepts but essential components of modern digital services.

Key Takeaways

- **Takeaway 1:** Definition of Data Pipelines.
- **Takeaway 2:** Core Components.
- **Takeaway 3:** Parallel Processing: